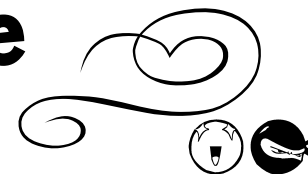




series and sequence

Part A: AP



2unit

NAME: _____

Q 1

ACE

The second term of an arithmetic series is 37 and the sixth term is 17.
What is the sum of the first ten terms?

- (A) 54
(B) 195
(C) 280
(D) 390

$$T_n = a + (n-1)d$$

$$T_2 = a + d = 37 \dots (1)$$

$$T_6 = a + 5d = 17 \dots (2)$$

$$\text{Eqn (2)} - (1)$$

$$4d = -20$$

$$d = -5$$

Substitute $d = -5$ into Eqn(1)

$$a - 5 = 37 \text{ or } a = 42$$

$$S_n = \frac{n}{2} [2a + (n-1)d]$$

$$= \frac{10}{2} [2 \times 42 + (10-1) \times -5]$$

$$= 195$$

1 Mark: B

Q 3

ASCHAM

The first and last terms of an arithmetic series are 10 and 60.
If the sum of the series is 3535, how many terms are there in the series?

(A) 11

(B) 101

(C) 110

(D) 51

Q 4

BARKER

Evaluate the arithmetic series $180 + 165 + 150 + \dots -15 -30$

$$12) a) a = 180 \quad d = -15$$

$$l = a + (n-1)d$$

$$-30 = 180 + (n-1) \times -15$$

$$-210 = -15n + 15$$

$$-15n = -225$$

$$n = 15$$

$$S_{15} = \frac{15}{2} (180 - 30)$$

$$= 1125$$

Q 2

BARKER

Evaluate $\sum_{r=3}^{10} 2r+1$.

(A) 120

(B) 91

(C) 122

(D) 112

$$\sum_{r=3}^{10} 2r+1 = 7+9+11+\dots+21$$

$$= 112$$

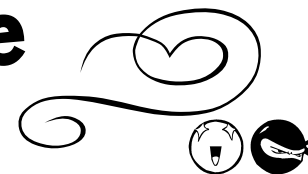
(D)





series and sequence

Part A: AP



Unit

NAME:

Q 5

Evaluate $\sum_{k=15}^{36} (3k-2)$.

CARINGBAH

$$\begin{aligned} c) & 43 + 46 + 49 + \dots + 106 \quad a=43, d=3 \\ S_{22} &= \frac{22}{2} [43 + 106] \\ &= \underline{1639} \end{aligned}$$

Q 6

The first three terms of an arithmetic sequence are $-1 + 4 + 9 + \dots$

- Find the 50th term.
- Find the sum of the first fifty terms.

CARINGBAH

f) $a = -1 \quad d = 5$

$$\begin{aligned} i) T_{50} &= a + 49d \\ &= -1 + 49(5) \\ &= 244 \\ ii) S_{50} &= \frac{50}{2}(a + L) \\ &= 25(-1 + 244) \\ &= 6075 \end{aligned}$$

2

Q 7

The first and last terms of an arithmetic series are 10 and 60.
If the sum of the series is 3535, how many terms are there in the series?

- (A) 11 (B) 101 (C) 110 (D) 51

A
S
C
H
A
M

Q 8

Evaluate $\sum_{n=2}^{16} (13-5n)$.

CSSA

1

1

$$\begin{aligned} \sum_{n=2}^{16} 13 - 5n &= (13-10) + (13-15) + (13-20) + \dots + (13-80) \\ &= 3 - 2 - 7 - \dots - 67 \end{aligned}$$

This is an arithmetic series with $a = 3$, $d = -5$ and $n = 15$.

$$\begin{aligned} S_n &= \frac{n}{2}(2a + (n-1)d) & S_n &= \frac{n}{2}(a + l) \\ &= \frac{15}{2}(2 \times 3 + 14 \times (-5)) & \text{or} & S_n = \frac{15}{2}(3 - 67) \\ &= -480 & & = -480 \end{aligned}$$



series and sequence

Part A: AP

2unit

NAME: _____

Q 9

FORT ST

An arithmetic progression has a first term 1 and last term 14.
The sum of the series is 90.

- (i) Find the number of terms in the series.
(ii) Show that the common difference is $\frac{13}{11}$.

$$AP: a = 1 \quad L = 14 \quad S_n = 90$$

$$S_n = \frac{n}{2}(a + L)$$

$$90 = \frac{n}{2}(1 + 14) \quad \checkmark$$

$$180 = 15n$$

$$n = 12 \quad \checkmark$$

12 terms in the series.

$$S_n = \frac{n}{2}[2a + (n-1)d]$$

$$90 = \frac{6}{2}[2 + (12-1)d] \quad \checkmark$$

$$90 = 6(2 + 11d)$$

$$15 = 2 + 11d \quad \checkmark$$

$$11d = 13$$

$$d = \frac{13}{11} \text{ as req'd.}$$

Q10

FORT ST

The first three terms of an arithmetic series are 50, 43, 36.

- (i) Write down a formula for the n th term.
(ii) If the last term of the series is -27, how many terms are there in the series?
(iii) Find the sum of the series.

$$a_n = a + (n-1)d \quad (i) \quad 57 - 7n = -27$$

$$a = 50 \quad d = -7$$

$$t_n = 50 + (n-1)(-7) \quad \checkmark$$

$$= 50 - 7n + 7$$

$$= 57 - 7n$$

$$-7n = -84$$

$$n = 12 \quad \checkmark$$

There are 12 terms in the series

$$(iii) \quad S_n = \frac{n}{2}(a + L)$$

[or can use $S_n = \frac{n}{2}[2a + (n-1)d]$]

$$n = 12 \quad a = 50 \quad L = -27$$

$$S_{12} = \frac{12}{2}(50 - 27)$$

$$= 6 \times 23$$

$$= 138$$

Q11

FORT ST

An arithmetic series is given as $\sum_{n=1}^k 4n - 1$. What is the third term of this series?

- (A) $4k - 1$
(B) 3
(C) 11
(D) 21

Q12

GIRAWEN

The first and last terms of an arithmetic series are 10 and 60 respectively, and the sum of the series is 3535. Find: (i) the number of terms in the series.
(ii) the common difference.

(a)(i) Arithmetic series: $a = 10, L = 60$
 $S_n = 3535$

$$S_n = \frac{n}{2}(a + L)$$

$$3535 = \frac{n}{2}(10 + 60)$$

$$35n = 3535$$

$$n = \frac{3535}{35}$$

$$\therefore n = 101 \quad (2m)$$

(a)(ii) $L = a + (n-1)d$

$$60 = 10 + 100 \times d$$

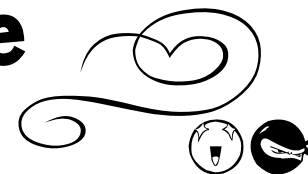
$$100d = 50$$

$$\therefore d = \frac{1}{2} \quad (2m)$$



series and sequence

Part A: AP



2unit

NAME: _____

Q13

An arithmetic sequence begins with the three terms $-6, 1, 8$.

GIRAWEEEN

- Find the 200th term of the sequence.
- Find the sum of 200 terms of the sequence.

$$(a) a = -6 \quad d = 7$$

$$(1) T_n = a + (n-1)d$$

$$T_{200} = -6 + 199 \times 7$$

$$T_{200} = 1387$$

$$(11) S_n = \frac{n}{2} [2a + (n-1)d]$$

$$S_{200} = \frac{200}{2} [-12 + 199 \times 7]$$

$$S_{200} = 138100$$

Q14

The fifth term of an Arithmetic sequence is 14 and the sum of the first ten terms is 165.

Find the first term and the common difference of the sequence.

GOSFORD

$$T_5 = a + 4d = 14 \rightarrow a = 14 - 4d$$

$$S_{10} = 5(2a + 9d) = 165 \rightarrow 2a + 9d = 33$$

$$\therefore 2(14 - 4d) + 9d = 33$$

$$28 - 8d + 9d = 33$$

$$d = 5 \quad \therefore a = -6$$

Q15

For the sequence 100, 95, 90, 85.....

GOSFORD

- Show the sequence is Arithmetic.
- Find the 25th term.
- How many terms are needed to give a sum of zero?

$$14) a) 100, 95, 90, \dots$$

$$i) A.P. \text{ if } \dots$$

$$95 - 100 = 90 - 95$$

$$-5 = -5$$

True $\therefore$ A.P.

$$ii) T_{25} = 100 + (25-1) \times -5$$

$$= 100 - 120$$

$$= -20$$

$$iii) S_n = \frac{n}{2} (2a + (n-1)d)$$

$$0 = \frac{n}{2} (200 + (n-1)(-5))$$

$$0 = \frac{n}{2} (200 - 5n + 5)$$

$$0 = n(205 - 5n)$$

$$n = 0, 41$$

$n = 0$ not a solution

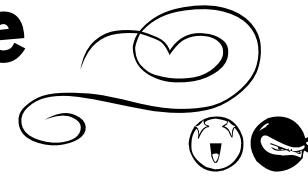
$\therefore$ 41 terms





series and sequence

Part A: AP



2unit

NAME: _____

Q16

GOSFROD

Consider the arithmetic sequence beginning with $-7, -3, 1 \dots$

- Find the 25th term in the sequence.
- Show that the sum of the first 20 terms is 620
- How many terms must be taken to give a sum of 221?

$$a = -7$$

$$d = 4$$

$$(i) T_n = a + (n-1)d$$

$$T_{25} = -7 + (25-1) \times 4$$

$$= 89$$

$$(ii) S_n = \frac{n}{2} [2a + (n-1)d]$$

$$S_{20} = \frac{20}{2} [2(-7) + (20-1)(4)]$$

$$= 620$$

$$(iii) S_n = \frac{n}{2} [2a + (n-1)d]$$

$$221 = \frac{n}{2} [2(-7) + (n-1)(4)]$$

$$442 = n(-14 + 4n - 4)$$

$$442 = 4n^2 - 18n$$

$$4n^2 - 18n - 442 = 0$$

$$2n^2 - 9n - 221 = 0$$

$$(n-13)(2n+17) = 0$$

$$n = 13, -\frac{17}{2}$$

$$\text{but } n > 0$$

$$\therefore n = 13$$

Q17

HORNSBY

The first three terms of an arithmetic series are 50, 43 and 36.

- Write down the n^{th} term of the series.
- If the last term of the series is -13 , how many terms are there in the series?
- Find the sum of the series.

$$b)_{ij} T_n = a + (n-1)d$$

$$= 50 + (n-1) \times -7$$

$$= 50 - 7n + 7$$

$$= 57 - 7n$$

$$ii) -13 = 57 - 7n$$

$$7n = 70$$

$$n = 10$$

$$iii) S_n = \frac{n}{2} (2a + (n-1)d)$$

$$S_{10} = \frac{10}{2} (2 \times 50 + (9 \times -7))$$

$$= 5(100 - 63)$$

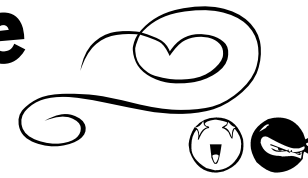
$$= 5 \times 37$$

$$= 185$$



series and sequence

Part A: AP



Unit

NAME: _____

Q18

HURLSTONE

The first three terms of an arithmetic sequence are 7, 11 and 15.

- Is 111 a term in this sequence? Justify your answer, by performing appropriate calculations.
- Find the sum of the first twenty-six terms.

a) (i)

$$a = 7, d = 4$$

$$T_n = a + (n-1)d$$

$$111 = 7 + (n-1)4$$

$$104 = 4n - 4$$

$$108 = 4n$$

$$n = 27$$

$\therefore$ yes 111 is the 27th term in this sequence.

(ii)

$$S_n = \frac{n}{2}(a+l) \quad l = 111 - 4 = 107$$

$$S_{26} = \frac{26}{2}(7+107) \\ = 1482$$

Q19

HURLSTONE

The fourth term of an arithmetic series is 27 and the seventh term is 12. What is the common difference?

(A) -5

(B) 5

(C) 13

(D) 42

Q20

HURLSTONE

The second term of an arithmetic series is 37 and the sixth term is 17.
What is the sum of the first ten terms of the series?

Common difference = -5

Therefore $T_1 = 42$ and $T_{10} = -3$

$$S_{10} = \frac{10}{2}(42-3) = 195$$

Q21

INTERNATIONAL GRAMMAR

For the arithmetic sequence 4, 9, 14, 19,

- Write the rule to describe the nth term.
- What is the 100th term?
- Find the sum of the first 100 terms.

$$4, 9, 14, 19.$$

$$i. \quad a = 4 \\ d = 5$$

$$T_n = a + (n-1)d \\ = 4 + n(5) - 4 \\ = 5n - 1 \quad \checkmark$$

$$ii. \quad T_{100} = 5 \times 100 - 1 \\ = 499 \quad \checkmark$$

$$iii. \quad S_{100} = \frac{n}{2}[2a + (n-1)d] \\ = \frac{100}{2}[2 \times 4 + (100-1) \times 5] \\ = 25150 \quad \checkmark$$



series and sequence

Part A: AP

2unit

NAME: _____

Q22

KNOX

Consider the series $\sqrt{5} + 3\sqrt{5} + 5\sqrt{5} + \dots = 225\sqrt{5}$

How many terms are in this series?

- (A) 15
(B) 16
(C) 113
(D) 225

Q23

MANLY

For $x > 1$, which one of the following expressions is the limiting sum of the given infinite series?

$$1 - \frac{1}{x} + \frac{1}{x^2} - \frac{1}{x^3} + \frac{1}{x^4} - \frac{1}{x^5} + \dots$$

- A $\frac{x}{1+x}$ B $\frac{x}{1-x}$ C $\frac{1+x}{x}$ D $\frac{1-x}{x}$

Q24

NEAP

(i) Show that $\sqrt{7} + \sqrt{28} + \sqrt{63} + \dots$ is an arithmetic series.

(ii) $\sqrt{7} + \sqrt{28} + \sqrt{63} + \dots + p = 300\sqrt{7}$.

How many terms are in the sequence?

(i) $\sqrt{7} + \sqrt{28} + \sqrt{63} \dots = \sqrt{7} + 2\sqrt{7} + 3\sqrt{7} \dots$
This is an arithmetic series as the common difference is $\sqrt{7}$.

(ii) $a = \sqrt{7}$ $d = \sqrt{7}$ $S_n = 300\sqrt{7}$

$$S_n = \frac{n}{2} \{2a + (n-1)d\}$$

$$300\sqrt{7} = \frac{n}{2} \{2\sqrt{7} + (n-1)\sqrt{7}\}$$

$$300\sqrt{7} = \frac{n}{2} (\sqrt{7} + n\sqrt{7})$$

$$300 = \frac{n}{2} (1+n)$$

$$600 = n + n^2$$

$$n^2 + n - 600 = 0$$

$$(n+25)(n-24) = 0$$

$$\therefore n = 24 \text{ as } n > 0$$

Q25

NORMANHURST

Boxes are stacked in layers, where each layer contains one box less than the layer below. There are six boxes in the top layer, seven in the next and so on. There are n layers altogether.

i. Write down the number of boxes in the bottom layer.

ii. Show that there are $\frac{1}{2}n(n+11)$ boxes.

i. (1 mark)

$$5 + n$$

ii. (2 marks)

$$\begin{aligned} S_n &= \frac{n}{2}(a + \ell) \\ &= \frac{n}{2}(6 + (5 + n)) \\ &= \frac{1}{2}n(11 + n) \end{aligned}$$

Q26

PENRITH

An arithmetic series has first term 3 and common difference 4.

a. Find the 17th term.

b. Find the sum of the tenth to the twentieth terms (inclusive).

B) i) a) $a=3$
 $d=4$

$$T_n = a + (n-1)d$$

$$T_{17} = 3 + (16)4$$

$$= 67$$

b) $S_{20} - S_{10} + T_{10}$

$$S_n = \frac{n}{2} [2a + (n-1)d]$$

$$S_{10} = 5(6 + (9)4)$$

$$= 210$$

$$S_{20} = 10(6 + (19)4)$$

$$= 820$$

$$T_{10} = 3 + (9)4$$

$$= 39$$

$$\therefore \text{sum} = 820 - 210 + 39$$

$$= 649$$



series and sequence

Part B: AP Problems

2unit

NAME: _____

Q 1

CARINGBAH

Penny decides to save money by putting some coins in a jar every day. On the first day she will put in 10c, on the second, 15c, and so on, with the amount she puts into the jar increasing by 5c each day.

- i) How much will she put in on the 15th day? 1
- ii) For how many days must she save before the jar holds \$37.00? 3

b) $a = 10$ $d = 5$ ii) $S_n = \frac{n}{2}(2a + (n-1)d)$

i) $T_{15} = a + 14d$ $3700 = \frac{n}{2}(20 + (n-1)5)$

$$= 10 + 14(5)$$

$$7400 = n(15 + 5n)$$

$$= 80c$$

$$5n^2 + 15n - 7400 = 0$$

$$n^2 + 3n - 1480 = 0$$

$$n = \frac{-3 \pm \sqrt{9 - 4(1)(-1480)}}{2(1)}$$

$$= 37 \quad (n > 0)$$

Q 2

CARINGBAH

In January 2010 Jane decided to start saving for a holiday when she finishes Year 12.

She put \$10 into a shoe box for safe keeping. The next month, she added to her savings by placing \$12 in the shoe box. Each month she put in \$2 more than the previous month. In October 2015, Jane will place her final amount into the box.

- (i) How much will Jane put in the shoe box in October 2015? 2
- (ii) Calculate the total amount of her savings. 1
(Note: Jane's shoe box does not offer any interest).

a) $\$10 + \$12 + \$14 + \dots$
 $a = 10 \quad d = 2 \quad n = 5 \times 12 + 10$
 $= 70$

i) $T_n = 10 + 69 \times 2$
 $= \$148$

ii) $S_n = \frac{70}{2}(10 + 148)$
 $= \$5530$



series and sequence

Part B: AP Problems

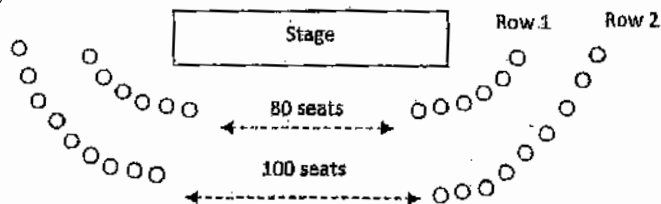
Unit

NAME: _____

Q 3

There are 80 seats in the first row of a concert hall. The next row has 100 and so on such that the next row has 20 more seats than the previous row.

CRANBROOK



Calculate:

- Find the n th term rule that will find how many seats are in each row. 1
- How many seats are there in the 10th row? 1
- If there 11100 seats in the concert hall, how many rows are there? 2

$$(i) T_1 = 80 = a$$

$$T_2 = 100$$

$$\therefore d = 20$$

$$\therefore T_n = a + (n-1)d$$

$$= 80 + (n-1) \times 20$$

$$= 60 + 20n$$

$$(ii) T_{10} = 60 + 20 \times 10 = 260$$

$$(iii) S_n = 11100 = \frac{n}{2} (2a + (n-1)d)$$

$$= \frac{n}{2} (160 + (n-1) \times 20)$$

$$22200 = n(140 + 20n)$$

$$11100 = n(7 + n)$$

$$0 = n^2 + 7n - 11100$$

$$= (n+37)(n-30)$$

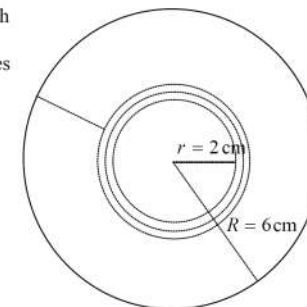
$$\therefore n = 30 - 37$$

There are 30 rows in the hall

Q 4

CSSA

The playing track of a CD is made out of a number of concentric circles with the inner circle having a radius of 2 cm and the outer circle having a radius of 6 cm. The CD is rotating at 5 revolutions per second and takes 25 minutes to completely play.



- Find the total number of revolutions for this CD.
- Find the total length of the playing track in km, correct to one decimal place.

Record is played at 5 revolutions per second for 25 minutes therefore will rotate

$$5 \times 60 \times 25 = 7500 \text{ revolutions}$$

Therefore there will be 7500 concentric circles whose radii increases in an arithmetic sequence from 2 cm to 6 cm. The length of the track record can be found using the sum of the arithmetic sequence with 7500 terms:

$$S_{7500} = \frac{7500}{2} (2\pi \times 2 + 2\pi \times 6) = 188495.5595 \text{ cm} = 1.9 \text{ km (1 d.p.)}$$



series and sequence

Part B: AP Problems

Unit

NAME: _____

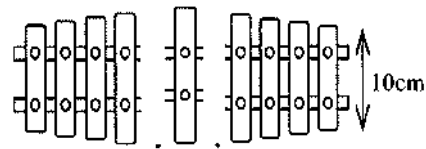
Q 7

CSSA

A percussionist is experimenting with designs for a xylophone. It is to be **symmetrical** in shape as shown in the diagram above.

The shortest wooden bar is to be 10cm long and the consecutive bars will differ in length by b cm. The total length of all the wooden bars is S cm.

Let the number of wooden bars be $2n+1$.



(i) Show that $S = bn^2 + 20n + 10$.

(ii) Given that $S = 360$ cm and $b = 1.5$ cm find the number of wooden bars.

Each side of the xylophone has n wooden bars and there is the extra one in the middle.
Treat this as two series:

$$\begin{aligned} S &= \frac{n}{2} [20 + (n-1)b] + \frac{n+1}{2} [20 + nb] \\ &= 10n + \frac{bn^2}{2} - \frac{nb}{2} + 10n + 10 + \frac{bn^2}{2} + \frac{nb}{2} \\ &= 20n + bn^2 + 10, \text{ as required} \end{aligned}$$

(a)(ii) (2 marks)

Sample answer

$$360 = \frac{3}{2}n^2 + 20n + 10$$

$$3n^2 + 40n - 700 = 0 \quad \therefore (3n+70)(n-10) = 0$$

$$n = 10, \quad n > 0$$

$$\text{Number of wooden bars} = 2(10) + 1 = 21$$

Q 8

FORT ST

Maureen is raising money for charity by jumping on a Pogo Stick. Her challenge is to jump between two points, A and B, 20 times. On her first attempt she takes 45 jumps. On her second attempt she takes 48 jumps. On her third attempt she takes 51 jumps. She continues this pattern for all 20 attempts.

- How many jumps did she make on her 20th attempt?
- How many jumps did she make altogether?

i.

$$AP: a = 45, d = 3$$

$$T_n = a + (n-1)d$$

$$T_1 = 45 \quad \therefore a = 45$$

$$T_2 = 48$$

$$T_3 = 51 \quad \therefore d = 3 \quad \text{for correct } a \text{ and } d$$

$$\therefore T_{20} = 45 + (20-1) \times 3 = 102 \text{ jumps}$$

ii.

$$\begin{aligned} S_{20} &= \frac{n}{2} (2a + (n-1)d) \\ &= \frac{20}{2} (2 \times 45 + 3 \times 19) \\ &= 1470 \text{ jumps} \end{aligned}$$



series and sequence

Part B: AP Problems

Unit

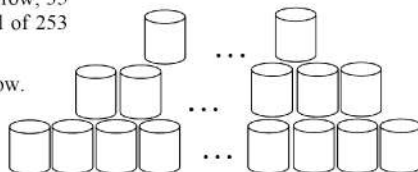
NAME: _____

Q 5

CSSA

Paint cans are stacked such that there are 38 cans on the bottom row, 35 cans on the next row, 32 on the next row and so on until a total of 253 cans are stacked.

- Write down a formula for the number of cans in the n th row.
- How many rows are there in this stack?
- How many cans are there in the final row of this stack?



Sample Answer

Using the formula $T_n = a + (n-1)d$, where $a = 38$ and $d = -3$:

$$\begin{aligned} T_n &= 38 + (n-1) \times -3 \\ &= 38 - 3n + 3 \\ &= 41 - 3n \end{aligned}$$

(a) (ii) (2 marks)

Substituting into the formula $S_n = \frac{n}{2}[2a + (n-1)d]$, where $S_n = 253$, $a = 38$ and $d = -3$:

$$253 = \frac{n}{2}[2 \times 38 + (n-1) \times -3]$$

$$506 = n(79 - 3n)$$

$$3n^2 - 79n + 506 = 0$$

Using the quadratic formula:

$$n = \frac{79 \pm \sqrt{6241 - 6072}}{6}$$

$$= 15.3 \text{ or } 11$$

$\therefore$ there are 11 rows of paint tins.

Q 6

CSSA

Each hour, a grandfather clock chimes the number of times that corresponds with the time of day.
How many times does the clock chime in one day?

$$2(1 + 2 + 3 + \dots + 12), \text{ where } n = 12, a = 1, l = 12$$

$$S_n = \frac{n}{2}(a + l)$$

$$S_{12} = 2 \times \left[\frac{12}{2}(1 + 12) \right] = 156$$





series and sequence

Part B: AP Problems

Unit

NAME: _____

Q 9

GIRAWEEEN

Jasmine sets aside savings each week for a vacation. In the first week she saves \$36 and each subsequent week she is able to save \$4 more than the previous week.

The maximum Jasmine is able to save is \$180 per week.

- How many weeks would it take for Jasmine to be able to save \$180 per week?
- Jasmine needs to save \$7356 for her vacation.
How many weeks would it take for Jasmine to save the amount of her vacation?

(i) Solution

The series is given by $36 + 40 + 44 + \dots$

A.P. where $a = 36$, $d = 4$

$$T_n = a + (n-1)d$$

$$180 = 36 + (n-1)4$$

$$144 = 4n - 4$$

$$4n = 148$$

$$n = 37 \text{ weeks}$$

(ii) Solution

Since \$180 is the maximum Jasmine is able to save the series is

$$36 + 40 + 44 + \dots + 180 + 180 + 180 + 180 + \dots$$

We want the sum of this series to be \$7356 and finding how many extra \$180 savings this would take.

$$36 + 40 + 44 + \dots + 180 + 180n = 7356$$

$$\frac{37}{2}[36 + 180] + 180n = 7356$$

$$3996 + 180n = 7356$$

$$180n = 3360$$

$$n = 19$$

$$\begin{aligned} \text{Therefore the total time} &= 37 + 19 \\ &= 56 \text{ weeks} \end{aligned}$$

Q10

TGHS

Madison is learning to drive. Her first lesson is 10 minutes long. Her second lesson is 15 minutes long. Each subsequent lesson is 5 minutes longer than the previous lesson.

- How long will Madison's fifteenth lesson be?
- How many minutes of lessons will Madison have completed after her fifteenth lesson?
- During which lesson will Madison have completed a total of 1150 minutes of driving lessons?

$$c) 10, 15, 20, \dots$$

$$i) \text{ AP with } a = 10, d = 5$$

$$\begin{aligned} T_{15} &= a + (n-1)d \\ &= 10 + 5(15-1) \\ &= 80 \text{ mins.} \end{aligned}$$

$$ii) \text{ i.e. Find } S_{15}$$

$$S_n = \frac{n}{2}(a + d)$$

$$\begin{aligned} S_{15} &= \frac{15}{2}(10 + 80) \\ &= 675 \text{ min} \\ &\rightarrow 11 \text{ h } 15 \text{ min} \end{aligned}$$

$$iii) \text{ i.e. Find } n \text{ when } S_n = 1150$$

$$S_n = \frac{n}{2}(2a + (n-1)d)$$

$$1150 = \frac{n}{2}(20 + 5(n-1))$$

$$= \frac{n}{2}(20 + 5n - 5)$$

$$= \frac{n}{2}(15 + 5n)$$

$$2300 = 15n + 5n^2$$

$$5n^2 + 15n - 2300 = 0$$

$$n^2 + 3n - 460 = 0$$

$$(n + 23)(n - 20) = 0$$

$$\therefore n = 20 \text{ (i.e. the 20th lesson)}$$



series and sequence

Part B: AP Problems

2 unit

NAME: _____

Q11

HORNSBY

For the last phase of preparations before the Olympic Games, swimmers started a new training schedule. On the first day they had to complete 26 laps of the pool. Each succeeding day they increased their training by 6 laps, until their daily schedule reached 200 laps. They then continued swimming 200 laps daily for a total of 15 days to fully complete their training schedule. (Note: length of pool = 50m)

- On which day did they first complete 200 laps?
- Find the total distance (in kilometres) completed by the swimmers.

$$26 + 32 + 38 + 44 + \dots + 200$$

(i) Arithmetic series $a = 26$

$$d = 6$$

$$T_n = 200$$

$$a + (n-1)d = 200$$

$$26 + 6(n-1) = 200$$

$$26 + 6n - 6 = 200$$

$$6n = 180$$

$$n = 30$$

They completed 200 laps on the 30th day

ii) Total ^{laps} for first 30 days

$$= \frac{n}{2} [2a + l]$$

$$= \frac{30}{2} [26 + 200]$$

$$= 15 (226)$$

$$= 3390$$

Total laps for next 15 days

$$= 15 \times 200$$

$$= 3000 \text{ laps.}$$

$$\text{Total distance} = 6390 \times 50 \text{m}$$

$$= 319500 \text{m}$$

$$= 319.5 \text{ km}$$

Q12

HORNSBY

During the first week of January a salesperson made \$15 000 worth of sales.

In the second week of January she made \$18 000 worth of sales and each week thereafter she continued to increase her sales by \$3 000.

- Write a formula for how much she made in sales in the n^{th} week.
- How much did she make in sales in the last week of December

(Use 52 weeks in the year).

- How much did she have in total sales for the year?
- During which week of the year did her total sales exceed the \$1 million?

$$\begin{aligned} \text{i) } T_n &= 15000 + (n-1)3000 \\ &= 12000 + 3000n \end{aligned} \quad (1M)$$

$$\begin{aligned} \text{ii) } T_{52} &= 12000 + 3000 \times 52 \\ &= 168000 \quad \text{She made \$168 000 in sales. (1M)} \end{aligned}$$

$$\begin{aligned} \text{iii) } S_{52} &= \frac{52}{2} (15000 + 168000) \\ &= 4758000 \quad \text{Total sales were \$4 758 000 (1M)} \end{aligned}$$

$$\text{iv) } \frac{n}{2} (15000 + 12000 + 3000n) > 1000000$$

$$27000n + 3000n^2 > 2000000$$

$$3n^2 + 27n - 2000 > 0$$

$$\text{Consider: } 3n^2 + 27n - 2000 = 0$$

$$n = \frac{-27 \pm \sqrt{27^2 + 4 \times 3 \times 2000}}{6}$$

$$\approx 21.709 \dots \text{ since } n > 0 \quad (2M)$$

$\therefore$ During 22nd week sales would exceed the \$1 million.

$$\text{NB } \begin{cases} n=21 & S_{21} = 945000 \\ n=22 & S_{22} = 1023000 \end{cases}$$



series and sequence

Part B: AP Problems

2unit

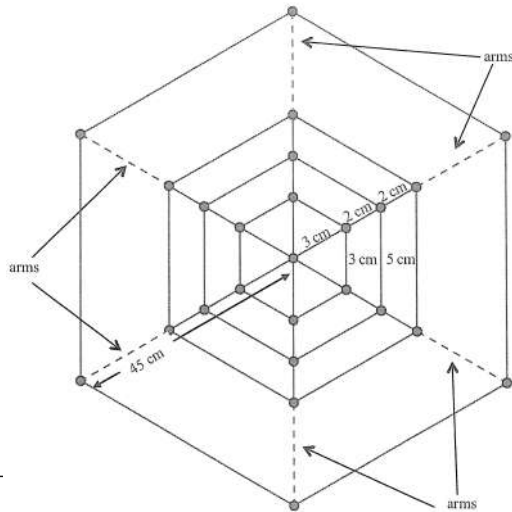
NAME: _____

Q13

HORNSBY

Incey Wincey spider makes a web in the shape of concentric regular hexagons. First he makes the 6 arms which are each 45 cm long. He then makes the sides of each regular hexagon. The first is 3 cm from the centre along each arm. Each successive regular hexagon is 2 cm further along the arm. The last hexagon is at the end of the arms.

- How many regular hexagons does Incey Wincey create?
- Find the total length of the web, including the arms, created by Incey Wincey.



(a)
(i) Change solutions
 $T_1 = 3$
 $T_2 = 5$
 $T_3 = 7$
 $a = 3, d = 2$
 $T_n = 3 + 2(n-1)$
 $= 3 + 2n - 2$
 $= 2n + 1$
 $2n + 1 = 45$
 $2n = 44$
 $n = 22$
 $\therefore 22$ regular hexagons

(ii)
Hexagon perimeter $= 6 \times 3 + 6 \times 5 + \dots + 6 \times 45$
 $= 6(3 + 5 + \dots + 45)$
 $= 6 \times \frac{22}{2} (3 + 45)$
 $= 66 \times 48$
 $= 3168$
 $= 3168 + 6 \times 45$
Total Length $= 3438 \text{ cm}$

Q14

HUNTERS

A sports club gained 50 members on its first day of operation. As the club's popularity grew, clientele grew steadily by an extra 14 members every subsequent day (ie 64 new members the second day, 78 new members on the third day and so on).

- How many members did the club have join on its 28th day of operation?
- After how many days of operation did the club have over 800 members join on one single day of operation?
- On which day did the 10 000th member join the club?

a) $50 + 64 + 78 + \dots$ Arithmetic Series

i) $T_n = a + (n-1)d$ ii) $T_n > 800$

$= 50 + 14(n-1)$ $36 + 14n > 800$

$= 50 + 14n - 14$ $n > 54.57$

$= 36 + 14n$

$T_{28} = 36 + 14(28)$

$= 428$

iii) $S_n = \frac{n}{2} (2a + (n-1)d)$

$10000 = \frac{n}{2} (100 + 14n - 14)$

$10000 = 43n + 7n^2$

$7n^2 + 43n - 10000 = 0$

$n = \frac{-43 \pm \sqrt{43^2 + (4 \times 7 \times 10000)}}{2 \times 7}$

$n = \frac{-43 + \sqrt{53089}}{14}$

$n = 34.84 \therefore \text{on the 35th Day}$





series and sequence

Part B: AP Problems

2 unit

NAME: _____

Q15

INDEPENDENT

- Farmer Brown has hired a driller to drill a borehole to enable her to have access to the underground water on her property. The driller quotes a price of \$260 for the first 3 metres drilled, \$280 for the next 2 metres, \$300 for the next 2 metres and so on. The price increases by the same amount for each successive 2 metres of borehole drilled.
- Show that the cost of drilling the portion from a depth of 25 metres to 27 metres is \$500.
 - Calculate the total cost of drilling to a depth of 27 metres.
 - The cost of drilling the borehole to reach water was \$12500. Find the total depth drilled to give access to the water.

$$\begin{aligned} \text{(b)(i) AP } a &= 3 \quad d = 2 \\ T_n &= a + (n-1)d \\ 27 &= 3 + (n-1)2 \\ n &= 13 \\ \text{Cost AP } a &= 260 \quad d = 20 \\ T_{13} &= 260 + (13-1)20 \\ &= \$500 \end{aligned}$$

$$\begin{aligned} \text{Q15(b)(ii)} \\ \text{Cost AP} \\ S_n &= \frac{n}{2}(2(260) + (n-1)20) \\ &= \$4940 \\ \text{(iii)} \\ 12500 &= \frac{n}{2}(2(260) + (n-1)20) \\ 12500 &= n^2 + 25n \\ n^2 + 25n - 12500 &= 0 \\ (n+50)(n-25) &= 0 \\ n &= 25 \quad (n \neq -25) \\ \text{Borehole AP} \\ T_{25} &= 3 + (25-1)2 \\ &= 51 \text{ metres} \end{aligned}$$

Q16

INDEPENDENT

- During the drought of the last few years, the water level in the local dam in the township of Wallaville was reduced to 2.5% of its capacity. In the first week of drought breaking rain, the inflow added 3% of capacity to the amount of water in the dam (i.e. the dam was 5.5% full). In the next week the inflow added 3.5% of capacity to the amount of water in the dam. In the third week 4% of capacity was added. This pattern continued so that each week an extra 0.5% of capacity was added to the dam until it was full.
- What percentage of capacity was added to the dam in the 10th week?
 - What percentage of capacity was in the dam after 10 weeks?
 - How many weeks would it have taken to fill the dam?

- (i) Amounts of inflows each week: $3 + 3.5 + 4 + \dots$. This is an Arithmetic series ($a = 3.5$, $d = 0.5$). The initial 2.5% is a separate term.

$$T_{10} = 3 + (10-1)0.5 = 7.5\% \text{ added in 10th week (1 mark)}$$
- (ii) $S_{10} = \frac{10}{2}[2 \times 3 + (10-1) \times 0.5]$ (1 mark)

= 52.5% inflow in 10 weeks

Dam holds $52.5\% + 2.5\% = 55\%$ (1 mark)
- (iii) When 100% full, 97.5% has been added (in n weeks).

$$S_n = \frac{n}{2}[2 \times 3 + (n-1) \times 0.5]$$

$$97.5 = \frac{n}{2}(0.5n + 5.5) \quad (1 \text{ mark})$$

$$390 = n^2 + 11n$$

$$(n-15)(n+26) = 0$$

$$n = 15 \quad (n \neq -26)$$

It took 15 weeks to fill the dam (1 mark)





series and sequence

Part B: AP Problems

2 unit

NAME: _____

Q17

INDEPENDENT

Peta is saving to buy a new car. She needs \$12 700 for the car. The first month she saves \$25. The next month she saves \$40. The following month she saves \$55.

If she continues to increase the amount she saves by \$15 each month how many months does it take Peta to save for her car?

$$S = 25 + 40 + 55 + \dots = \$12700$$

A.P. $a = 25$, $d = 15$ and $S_n = 12700$. Find n .

$$12700 = \frac{n}{2}(50 + (n-1) \times 15)$$

$$25400 = n(50 + 15n - 15)$$

$$= n(35 + 15n)$$

$$5080 = 7n + 3n^2$$

$$3n^2 + 7n - 5080 = 0$$

$$n = \frac{-7 \pm \sqrt{49 - 4 \times 3 \times -5080}}{6}$$

$$= \frac{-7 \pm \sqrt{61009}}{6}$$

$$= \frac{-7 \pm 247}{6}$$

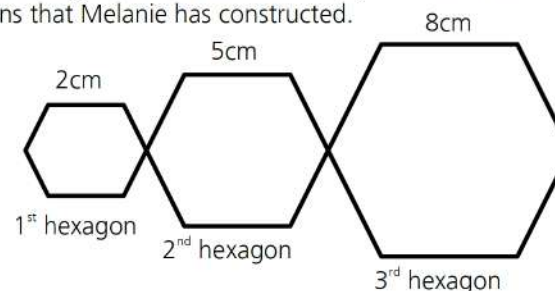
$$= 40 \text{ months (ignore negative answer)}$$

Q18

INTERNATIONAL GRAMMAR

Melanie is using wire to construct a geometrical design which consists of n regular hexagons with sides 2cm, 5cm, 8cm and so on going up by the same amount each hexagon. The diagram below shows the first 3 hexagons of her design.

- Find the perimeter of the n th hexagon.
- Show that the total length of the wire is $L = 9n^2 + 3n$.
- If the total length of the wire is 6 metres, find the number of hexagons that Melanie has constructed.



$$\begin{aligned} \text{a) } P &= 6(2 + (n-1) \times 3) \\ &= 6(2 + 3n - 3) \\ &= 18n - 6 \end{aligned}$$

$$\begin{aligned} \text{ii) } S_n &= \frac{n}{2}(18n - 6 + 2 \times 6) \\ &= \frac{n}{2}(18n + 6) \\ &= 9n^2 + 3n \end{aligned}$$

$$\text{iii) } 600 = 9n^2 + 3n$$

$$\begin{aligned} 3n^2 + n - 200 &= 0 \\ (3n + 25)(n - 8) &= 0 \end{aligned}$$

$$n = -\frac{25}{3}, 8$$

$$n = 8$$



series and sequence

PART B: AP PROBLEMS

2unit

NAME: _____

Q19

JAMES RUSE

A polygon has 60 sides. The lengths of the sides, starting with the smallest, form an arithmetic sequence. The perimeter of the polygon is 1770 cms and the length of the longest side is three times the length of the shortest side. Find the lengths of the two shortest sides.

(c) let a be the first term and d the common difference

$$S_{60} = \frac{60}{2} [2a + 59d] = 1770$$

$$6a + 177d = 177 \quad \dots\dots (1)$$

Also $a + 59d = 3a$
 $2a = 59d$
 $\therefore 6a = 177d \quad \dots\dots (2)$
 Sub (2) in (1)

$$177d + 177d = 177 \quad \therefore d = \frac{1}{2}$$

$\therefore$ lengths of shortest sides are $14\frac{3}{4}$, $15\frac{1}{4}$ cm

Q20

NEAP

1st

•

2nd

• •

3rd

• • •
• • •

4th

• • • •
• • • •
• • • •

5th

• • • • •
• • • • •
• • • • •
• • • • •

$$1 + 2$$

$$= 3$$

$$1 + 2 + 3$$

$$= 6$$

$$1 + 2 + 3 + 4$$

$$= 10$$

$$1 + 2 + 3 + 4 + 5$$

$$= 15$$

The diagram shows the first five triangular numbers. Calculate the value of the 78th triangular number.

$$T_{78} = 1 + 2 + 3 + 4 + \dots + 78$$

$$T_{78} = \frac{78}{2} \{2 \times 1 + (78 - 1) \times 1\}$$

$$= 3081$$



series and sequence

Part B: AP Problems

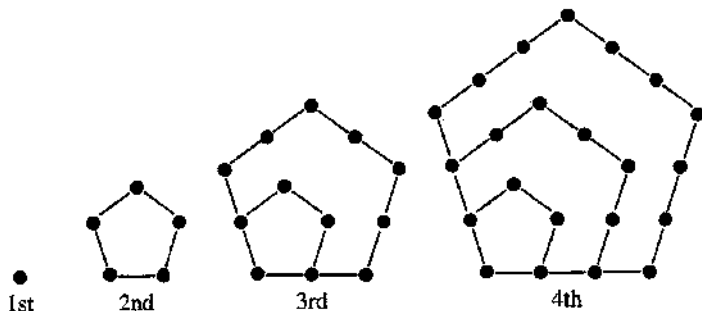
2 unit

NAME: _____

Q21

NEAP

Josh is making a pentagonal dot pattern.



The table below shows the number of dots on the first four patterns.

Diagram	1st	2nd	3rd	4th
Number of dots on the bottom row (n)	1	2	3	4
Total number of dots in the diagram (T)	1	5	12	22
Additional dots added to the previous pattern to make this pattern	1	4	7	10

- (i) How many dots will Josh have to add to the 17th diagram to make the 18th diagram?
 (ii) How many dots are in the 18th diagram?

(i) $T_n = a + (n-1)d$ of 1, 4, 7, 10...

$$T_{18} = 1 + (18-1) \times 3$$

$$T_{18} = 52$$

(ii) $S_n = \frac{n}{2} \{2a + (n-1)d\}$

$$\therefore S_{18} = \frac{18}{2} \{2 + 17 \times 3\}$$

$$S_{18} = 477$$

There are 477 dots in the 18th diagram.

Q22

NORMANHURST

Damien runs a marathon, which is a distance of 42 km. He runs the first kilometre in 4 minutes, and from there on, he takes 5 seconds longer to run each successive kilometre.
 (i.e the 2nd kilometre takes 4 minutes and 5 seconds.)

- (i) How long does he take to run the 16th kilometre?
 (ii) How long does it take to run the entire marathon?
 (iii) How far had he run after 2 hours 36 minutes and 15 seconds?

a) This is an Arithmetic series with

(i) $a = 4 \times 60 = 240$, $d = 5$

Want the 16th term

$$u_n = a + (n-1)d$$

$$= 240 + 15 \times 5$$

$$= 315 \text{ sec}$$

$$= 5 \text{ min and } 15 \text{ sec}$$

a) (ii) Want the sum of the 42 km.

$$s_n = \frac{n}{2} (2a + (n-1)d)$$

$$= 21(480 + 41 \times 5)$$

$$= 14\,385 \text{ sec}$$

$$= 3 \text{ hrs } 59 \text{ min } 45 \text{ sec}$$

a) (iii) 2 hours 36 minutes and 15 seconds = 9375 sec

$$s_n = \frac{n}{2} (2a + (n-1)d)$$

$$9375 = \frac{n}{2} (480 + 5(n-1))$$

$$18750 = n(475 + 5n)$$

$$5n^2 + 475n - 18750 = 0$$

$$n^2 + 95n - 3750 = 0$$

$$n = \frac{-95 \pm \sqrt{9025 + 15000}}{2}$$

$$= \frac{-95 \pm 155}{2}$$

$$= 30 \text{ or } -125$$

Ignore the negative, so he has travelled 30 km.



series and sequence

Part B: AP Problems

Unit

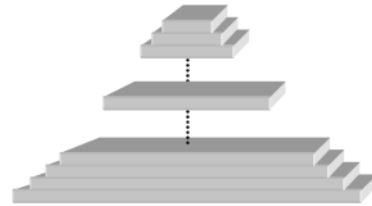
NAME: _____

Q23

NSGHS

A pyramid is built using 1536 blocks on the base level. The next layer contains 1472 blocks and the next 1408, and so on.

- How many blocks are used for the ninth layer?
- Before it is capped with a single pyramid block, the top layer has 64 blocks. How many layers are there before the cap is put on?
- How many blocks were used in the construction of the pyramid?



$$b) 1536 + 1472 + 1408 + \dots$$

$$a = 1536; d = -64$$

$$T_n = 1536 - 64(n-1)$$

$$i) T_9 = 1536 - 64(8)$$

$$= 1104$$

$$ii) 64 = 1536 - 64(n-1)$$

$$64(n-1) = 1472$$

$$n-1 = 23$$

$$\therefore n = 24 \text{ so } 24 \text{ layers.}$$

$$iii) S_n = \frac{n}{2}(a+l)$$

$$S_{24} = \frac{24}{2}(1536 + 64) + 1$$

$$= 19201 \text{ blocks.}$$

Q24

NSGHS

Con the fruiterer decides to display the oranges in the shape of a square-based pyramid. He has built a frame in the shape of a pyramid and begins by placing a row of oranges around the base.

In subsequent rows, each orange rests only on two oranges in the row below and on the frame. Con completes the pyramid by placing the last orange on top of the four previous oranges.



- Explain why each row of the pyramid excluding the last orange has 4 fewer oranges than the row on which it rests.
- Con uses 56 oranges in the first row. How many rows of oranges will he have placed *before* he places the last orange?
- How many oranges will Con use in completing his display?

- Each face has one less orange than in the row below. There are 4 faces. $\therefore$ there are 4 fewer oranges than the row on which it rests.

$$ii) 56 + 52 + 48 + \dots + 4$$

$$n = ? \quad T_n = a + (n-1)d$$

$$a = 56 \quad 4 = 56 + (n-1)(-4)$$

$$T_n = 4 \quad 4 = 56 - 4n + 4$$

$$d = -4 \quad 4n = 56$$

$$n = 14$$

$\therefore$ there are 14 rows

$$iii) S_n = \frac{n}{2}[a+l] + 1$$

$$= \frac{14}{2}[56+4] + 1$$

$$= 421$$

$$\therefore \text{he will use 421 oranges}$$



series and sequence

PART B: AP PROBLEMS

2unit

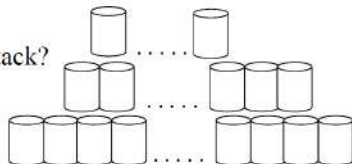
NAME: _____

Q25

NSGHS

Food tins are stacked so there are 49 tins on the bottom row, 45 tins on the next row, 41 tins on the row after and so on until a total of 321 cans are stacked.

- Write down a formula for the number of cans in the n^{th} row.
- How many rows are in the stack in total?
- How many cans are in the top row of this stack?



$$b) T_1 = 49, T_2 = 45, \dots$$

$$a = 49, d = -4$$

$$T_n = a + (n-1)d = 49 - 4(n-1) \checkmark$$

$$i) T_n = 53 - 4n \text{ OR } \checkmark$$

$$i) 321 = \frac{n}{2} (2a + (n-1)d)$$

$$642 = n(98 + (n-1) \times -4)$$

$$4n^2 - 102n + 642 = 0 \checkmark$$

$$n = \frac{51 \pm \sqrt{33}}{4}$$

$$= 14.18 \text{ or } 11.3$$

$n = 14.18$ is not the valid answer as it exceeds 321 tiles.

$\therefore n = 11.3$ is the correct answer $\checkmark$

Therefore, $n = 14$

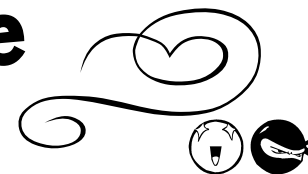
iii) 2 tiles on the TOP row.

most of them were given but they took it off!



series and sequence

Part C: GP



Unit

NAME: _____

Q 1

BARKER

(i) For what values of x can the geometric series $1 + 2x + 4x^2 + 8x^3 + \dots$ have a limiting sum?

(ii) For what values of x does $\sum_{r=0}^{\infty} (2x)^r = \frac{5}{9}$?

a) (i) $1 + 2x + 4x^2 + 8x^3 + \dots$
 $a = 1$ $r = 2x$
 For a limiting sum $-1 < r < 1$ $r \neq 0$
 $\therefore -1 < 2x < 1$
 $-\frac{1}{2} < x < \frac{1}{2}$ $x \neq 0$

(ii) $S_{\infty} = \frac{a}{1-r}$
 $\frac{5}{9} = \frac{1}{1-2x}$
 $5(1-2x) = 9$
 $5 - 10x = 9$
 $-10x = 4$
 $x = -\frac{2}{5}$

Q 2

BARKER

The limiting sum of the series $1 - \frac{1}{2} + \frac{1}{4} - \frac{1}{8} + \frac{1}{16} - \dots$ is

- (A) $\frac{1}{2}$ (B) $\frac{2}{3}$ (C) $1\frac{1}{2}$ (D) 2

$a = 1$ $r = -\frac{1}{2}$ $S_{\infty} = \frac{1}{1 - (-\frac{1}{2})} = \frac{1}{3/2} = \frac{2}{3}$
 B

Q 3

ACE

An infinite geometric series has a first term of 12 and a limiting sum of 15. What is the common ratio?

- (A) $\frac{1}{5}$
 (B) $\frac{1}{4}$
 (C) $\frac{1}{3}$
 (D) $\frac{1}{2}$

$a = 12$ and $S = 15$

$S = \frac{a}{1-r}$

$15 = \frac{12}{1-r}$

$15 - 15r = 12$

$15r = 3$

$r = \frac{1}{5}$

1 Mark: A

Q 4

BAULKHAM

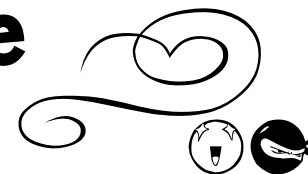
The sum of the series $\frac{4}{9} + \frac{2}{9} + \frac{1}{9} + \frac{1}{18} + \frac{1}{36} + \frac{1}{72} + \dots + \frac{1}{144}$ is

- (A) $\frac{127}{36}$ (B) $\frac{127}{81}$ (C) $\frac{127}{144}$ (D) $\frac{81}{127}$



series and sequence

Part C: gp



Unit

NAME: _____

Q 5

If the limiting sum of the series $1 - 2p + 4p^2 \dots$ is $\frac{4}{7}$. Find the value of p .

BAULKHAM

$$1 - 2p + 4p^2 \dots = \frac{4}{7}$$

$$\frac{4}{7} = \frac{1}{1+2p} \quad \checkmark$$

$$4 + 8p = 7 \quad \checkmark$$

$$p = \frac{3}{8}$$

Q 6

Find the sum of the first 15 terms of the series

$$1 + 3 + 3^2 + 3^3 + \dots$$

BAULKHAM

$$a=1, r=3, n=15 \quad \checkmark \quad s_n = \frac{a(r^n - 1)}{r - 1}$$

$$s_{15} = \frac{1(3^{15} - 1)}{3 - 1} = 7174453 \quad \checkmark$$

2

Q 7

A series is defined as $(\sqrt{2}-1) + (\sqrt{2}-1)^2 + (\sqrt{2}-1)^3 + \dots$

Find its limiting sum.

CRANBROOK

$$a = \sqrt{2} - 1$$

$$r = \sqrt{2} - 1$$

$$|r| = |\sqrt{2} - 1| < 1 \quad \therefore \text{Limiting sum exists}$$

$$S_{\infty} = \frac{a}{1-r} = \frac{\sqrt{2}-1}{1-(\sqrt{2}-1)} = \frac{\sqrt{2}-1}{2-\sqrt{2}} = \frac{(\sqrt{2}-1)(\sqrt{2}+1)}{2-\sqrt{2}} = \frac{2-\sqrt{2}+\sqrt{2}-1}{2-\sqrt{2}} = \frac{1}{2-\sqrt{2}}$$

careful with
[1-(\sqrt{2}-1) \neq 1-\sqrt{2}-1]

3

Q 8

CSSA

(i) Write $0.7\dot{5}$ as the sum of an infinite geometric series.

(ii) Hence, express $0.7\dot{5}$ as a fraction.

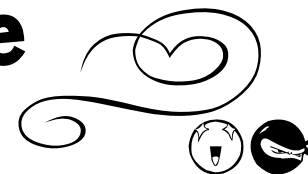
$$0.7\dot{5} = \frac{7}{10} + \frac{5}{100} + \frac{5}{1000} + \frac{5}{10000} \dots$$

$$S_{\infty} = \frac{a}{1-r} = \frac{7}{10} + \frac{\frac{5}{100}}{1 - \frac{1}{10}} = \frac{7}{10} + \frac{1}{18} = \frac{34}{45}$$



series and sequence

Part C: gp



Unit

NAME: _____

Q 9

CSSA

An infinite geometric series has a limiting sum of 24.
If the first term is 15, find the common ratio.

$$S_{\infty} = \frac{a}{1-r}$$

$$24 = \frac{15}{1-r}$$

$$24(1-r) = 15$$

$$r = \frac{3}{8}$$

Q10

CSSA

A series is defined as: $1, \frac{y^2}{(1-y)^2}, \frac{y^4}{(1-y)^4}, \dots$ $y \neq 0$ & $y \neq 1$

Determine the values of y so that the above series does NOT have an infinite sum.

The limiting sum does not exist if $|r| > 1$

$$\therefore \left| \frac{y^2}{(1-y)^2} \right| > 1 \quad \text{i.e.} \quad \frac{y^2}{(1-y)^2} > 1$$

$$\text{So } y^2 > (1-y)^2 \quad \therefore y^2 - (1-y)^2 > 0$$

$$\text{Expanding gives } y^2 - 1 + 2y - y^2 > 0$$

Q11

CSSA

What is the limiting sum of the geometric series

$$x^2 - 2x^3 + 4x^4 - 8x^5 + \dots \quad \text{where } |x| < 1?$$

(A) $\frac{x^2}{1+2x}$

(B) $\frac{x^2}{1-2x}$

(C) $\frac{1}{1+2x}$

(D) $\frac{x^2(1-(-2x)^n)}{1+2x}$

Q12

CSSA

Evaluate $\sum_{k=3}^{15} 2^k$

(A) $2^{15} - 2$

(C) $2^{15} - 8$

(B) $2^{13} - 8$

(D) $2^{16} - 8$



series and sequence

Part C: GP

Unit

NAME: _____

Q13

Consider the geometric series $1 + 3x + 9x^2 + 27x^3 + \dots$

FORT ST

- (i) For what values of x does this series have a limiting sum?
(ii) The limiting sum of this series is 100. Find the value of x

(i) For limiting sum
 $-1 < 3x < 1$ ✓
 $-\frac{1}{3} < x < \frac{1}{3}$ ✓

(ii) $S_{\infty} = \frac{a}{1-r}$
 $100 = \frac{1}{1-3x}$ ✓
 $1-3x = \frac{1}{100}$
 $-3x = -\frac{99}{100}$
 $x = \frac{33}{100}$ ✓

Q14

The third and seventh terms of a geometric series are 1.25 and 20 respectively. What is the first term?

FORT ST

- (A) ± 2
(B) ± 4

(C) $\frac{5}{16}$

(D) $\frac{5}{1024}$

$$T_3 = ar^2 = 1.25 \text{ and}$$

$$T_7 = ar^6 = 20$$

Divide the two equations $\frac{ar^6}{ar^2} = \frac{20}{1.25}$
 $r^4 = 16$
 $r = \pm 2$

$$T_7 = a \times (\pm 2)^6 = 20$$

$$a = \frac{20}{64} = \frac{5}{16}$$

Q15

An infinite geometric series has a first term of 3 and a limiting sum of 1.8. What is the common ratio?

FORT ST

- (A) -0.3
(B) -0.6

- (C) -1.5
(D) -3.75

$$a = 3 \text{ and } S = 1.8$$

$$S = \frac{a}{1-r}$$

$$1.8 = \frac{3}{1-r}$$

$$1.8 - 1.8r = 3$$

$$1.8r = -1.2$$

$$r = -0.6$$

Q16

GIRAWEN

Consider the geometric series $1 + (\sqrt{5} - 2) + (\sqrt{5} - 2)^2 + \dots$

- (i) Explain why the geometric series has a limiting sum.
(ii) Find the exact value of the limiting sum.

Write your answer with a rational denominator.

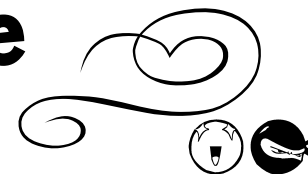
4(a)(i) $1 + (\sqrt{5}-2) + (\sqrt{5}-2)^2 + \dots$
 Add this is geometric series:
 have $a=1$, $r=\sqrt{5}-2$
 Limiting sum exists if $|r| < 1$
 in this case $r = \sqrt{5}-2 = 0.236\dots$
 i.e. $r < 1$
 $\therefore$ limiting sum exists. (1m)

4(a)(ii) $S_{\infty} = \frac{a}{1-r}$
 $= \frac{1}{1-(\sqrt{5}-2)}$
 $= \frac{1}{3-\sqrt{5}}$
 $= \frac{1}{3-\sqrt{5}} \times \frac{3+\sqrt{5}}{3+\sqrt{5}}$
 $= \frac{3+\sqrt{5}}{9-5}$
 $\therefore S_{\infty} = \frac{3+\sqrt{5}}{4}$



series and sequence

Part C: gp



Unit

NAME: _____

Q17

A geometric sequence begins with the three terms $-4, 8, -16$. Find the 15th term of the progression.

GIRAWEEEN

$$T_n = ar^{n-1}$$

$$T_{15} = -4(-2)^{14}$$

$$T_{15} = \underline{-65536}$$

Q18

In a geometric sequence the third term is 18 and the seventh term is 1458. Find the first term and the common ratio.

FORT ST

$$T_3 = ar^2$$

$$18 = ar^2$$

$$T_7 = ar^6$$

$$1458 = ar^6$$

$$\frac{1458}{18} = \frac{ar^6}{ar^2}$$

$$81 = r^4$$

$$r = \pm 3$$

$$ar^2 = 18$$

$$a \times 9 = 18$$

$$a = 2$$

Q19

The limiting sum of the series $x + x^2 + \dots$ is 15. If $|x| < 1$, the value of x is

- (A) $-\frac{16}{15}$ (B) $-\frac{15}{16}$ (C) $\frac{16}{15}$ (D) $\frac{15}{16}$

HUNTERS

$$S_{\infty} = \frac{a}{1-r}$$

$$\frac{x}{1-x} = 15$$

$$x = 15(1-x)$$

$$16x = 15$$

$$x = \frac{15}{16}$$

Q20

The first three terms of a geometric series are: $4a^2b^2, x, a^2 + 2ab + b^2$. Find x in terms of a and b .

HURLSTONE

Since the series is geometric:

$$\frac{T_1}{T_2} = \frac{T_2}{T_3}$$

$$\text{ie. } \frac{a^2 + 2ab + b^2}{x} = \frac{x}{4a^2b^2}$$

$$x^2 = 4a^2b^2(a^2 + 2ab + b^2)$$

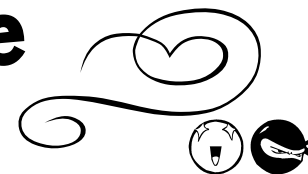
$$= 4a^2b^2(a+b)^2$$

$$x = \pm 2ab(a+b)$$



series and sequence

Part C: gp



2unit

NAME: _____

Q21

HURLSTONE

Consider the geometric series $1 + (\sqrt{11} - 3) + (\sqrt{11} - 3)^2 + \dots$

- (i) Explain why the geometric series has a limiting sum.
 (ii) Find the exact value of the limiting sum. Write your answer with a rational denominator.

(i)
 $r = \sqrt{11} - 3$
 ≈ 0.3166
 Since $-1 < r < 1$ the geometric series has a limiting sum.

(ii)

$$S = \frac{a}{1-r}$$

$$= \frac{1}{1 - (\sqrt{11} - 3)}$$

$$= \frac{1}{4 - \sqrt{11}} \times \frac{4 + \sqrt{11}}{4 + \sqrt{11}}$$

$$= \frac{4 + \sqrt{11}}{16 - 11}$$

$$= \frac{4 + \sqrt{11}}{5}$$

Q22

HURLSTONE

Consider the geometric series $5 + 10x + 20x^2 + 40x^3 + \dots$

- (i) For what values of x does this series have a limiting sum?
 (ii) The limiting sum of the series is 100. Find the value of x .

(e)
 (i) limiting sum if $|r| < 1$
 $r = \frac{10x}{5} = \frac{20x^2}{10x}$
 $r = 2x$
 $|2x| < 1$
 $2x < 1$ or $-2x < 1$
 $x < \frac{1}{2}$ or $x > -\frac{1}{2}$
 $\therefore -\frac{1}{2} < x < \frac{1}{2}$

(ii)
 Limiting sum:
 $S_{\infty} = \frac{a}{1-r}$
 $100 = \frac{5}{1-2x}$
 $100 - 200x = 5$
 $200x = 95$
 $x = \frac{19}{40}$ or 0.475

Q23

INDEPENDENT

The geometric series $1 - x + x^2 - \dots$ has a limiting sum of 4.
 Find the value of x .

$$a = 1, r = -x$$

$$4 = \frac{1}{1 - (-x)}$$

$$4 + 4x = 1$$

$$x = -\frac{3}{4}$$

Q24

INDEPENDENT

Evaluate: $\sum_{n=2}^4 3^{-n}$ (Give your answer in exact form).

$$3^{-2} + 3^{-3} + 3^{-4}$$

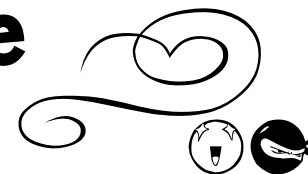
$$= \frac{1}{9} + \frac{1}{27} + \frac{1}{81}$$

$$= \frac{13}{81}$$



series and sequence

Part C: GP



Unit

NAME: _____

Q25

INDEPENDENT

Given the infinite series: $\frac{x}{3} + \frac{2x^2}{9} + \frac{4x^3}{27} + \dots$:

- Show that it is a geometric series.
- Find the values of x such that the series has a limiting sum and find the sum (in terms of x).

(c)(i) $a = \frac{x}{3}, r = \frac{2x}{3} = \frac{T_2}{T_1} = \frac{T_3}{T_2}$ (1 mark)

(c)(ii) Limiting Sum exists if $|r| < 1$ $-1 < \frac{2x}{3} < 1$
 $-\frac{3}{2} < x < \frac{3}{2}$ (1 mark)

$$S = \frac{a}{1-r} = \frac{\frac{x}{3}}{1-\frac{2x}{3}} = \frac{x}{3-2x} \quad (1 \text{ mark})$$

Q26

INDEPENDENT

Consider the geometric series $3 - 6y + 12y^2 - 24y^3 + \dots$

- For what values of y does this series have a limiting sum?
- The limiting sum of the series is $2\frac{1}{4}$. Find the value of y .

a) (i) 2 marks
Sample Answer:-

$$3 - 6y + 12y^2 - 24y^3 + \dots$$

$$r = -2y$$

$$|-2y| < 1$$

$$-1 < 2y < 1$$

$$-\frac{1}{2} < y < \frac{1}{2}$$

a) (ii) 2 marks
Sample Answer:-

$$S_{\infty} = \frac{3}{1+2y}$$

$$\frac{9}{4} = \frac{3}{1+2y}$$

$$9 + 18y = 12$$

$$18y = 3$$

$$y = \frac{1}{6}$$

Q27

INTERNATIONAL GRAMMAR

The third and seventh terms of a geometric series are 1.25 and 20 respectively. What is the first term?

$$T_3 = ar^2 = 1.25 \quad (1)$$

$$T_7 = ar^6 = 20 \quad (2)$$

$$(2) \div (1)$$

$$\frac{ar^6}{ar^2} = \frac{20}{1.25}$$

$$r^4 = 16$$

$$r = \pm 2$$

$$T_3 = ar^2$$

$$= a \times (\pm 2)^2 = 1.25$$

$$4a = 1.25$$

$$a = \frac{1.25}{4}$$

$$= \frac{5}{16} \text{ or } 0.3125 \checkmark$$

Q28

NEAP

A golf ball is dropped from a height of one metre. Each time it hits the ground it bounces to two-thirds of its previous height.

$$1 + 2 \times \left[\frac{2}{3} + \left(\frac{2}{3} \right)^2 + \dots \right]$$

$$= 1 + 2 \times \frac{a}{1-r}$$

$$= 1 + 2 \times \frac{\frac{2}{3}}{1-\frac{2}{3}}$$

$$= 1 + 2 \times \frac{2}{3} \times \frac{3}{1}$$

$$= 5 \text{ m}$$



series and sequence

Part C: GP

Unit

NAME: _____

Q29

NORMANHURST

(i) Write the first three terms of the series whose general term is given by $T_n = \frac{2^n}{3^{n-1}}$.

(ii) Evaluate $\sum_{n=1}^{\infty} \frac{2^n}{3^{n-1}}$.

$$\begin{aligned} (i) \\ T_1 &= 2 \\ T_2 &= \frac{4}{3} \\ T_3 &= \frac{8}{9} \\ \therefore 2, \frac{4}{3}, \frac{8}{9}, \dots \end{aligned}$$

$$\begin{aligned} (ii) \\ a &= 2 \\ \frac{T_2}{T_1} &= \frac{T_3}{T_2} = r \\ \frac{4}{3} \times \frac{1}{2} &= \frac{8}{9} \times \frac{3}{4} = \frac{2}{3} \\ \therefore r &= \frac{2}{3} \\ S_{\infty} &= \frac{2}{1 - \frac{2}{3}} = 6 \end{aligned}$$

Q30

NORTH SYDNEY BOYS

For the geometric series

$$1 - 3x + 9x^2 - \dots$$

i. find the values of x for which the limiting sum exists.

ii. find the value of x for which the limiting sum is $\frac{4}{5}$.

(i) Limiting sum exists when

$$\begin{aligned} -1 &< r < 1 \\ -1 &< -3x < 1 \\ 1 &> 3x > -1 \\ \therefore \frac{1}{3} &> x > -\frac{1}{3} \end{aligned}$$

$$\therefore \boxed{-\frac{1}{3} < x < \frac{1}{3}}$$

$$(ii) S_{\infty} = \frac{a}{1-r}$$

$$\frac{4}{5} = \frac{1}{1+3x}$$

$$4 + 12x = 5$$

$$12x = 1$$

$$x = \frac{1}{12}$$

Q31

PENRITH

The sum of the first 8 terms of a geometric series is 17 times the sum of its first 4 terms. Find the common ratio.

$$3 \text{ ii. } S_8 = 17S_4$$

$$\frac{r^8 - 1}{r - 1} = 17 \frac{r^4 - 1}{r - 1}$$

$$\begin{aligned} r^8 - 1 &= 17(r^4 - 1) \\ (r^4 - 1)(r^4 + 1) &= 17(r^4 - 1) \\ r^4 + 1 &= 17 \\ r^4 &= 16 \\ r &= \pm 2 \\ \therefore r &= 2 \end{aligned}$$

Q32

PLC

In a geometric series, all the terms are positive, the second term is 24 and the fourth term is

$$13\frac{1}{2}. \text{ Find}$$

(i) the first term,

(ii) the sum to infinity of the series.

$$c) i) T_2 = 24 \quad GP: T_n = ar^{n-1}$$

$$T_4 = 13\frac{1}{2}$$

$$ar = 24 \quad (1)$$

$$ar^3 = \frac{27}{2} \quad (2)$$

$$(2) \div (1) \quad r^2 = \frac{27}{48}$$

$$r = \pm \sqrt{\frac{27}{48}}$$

$$r = \pm \frac{3}{4} \quad (1 \text{ mark})$$

$$\text{since } r > 0 \text{ then } r = \frac{3}{4} \text{ and } a = \frac{24}{\frac{3}{4}} = 32 \quad (1)$$



series and sequence

Part D: GP Problems

2unit

NAME: _____

Q 1

ACE

Brendan planted a wattle tree three years ago when it was 70 cm tall. At the end of the first year after planting, the wattle tree was 120 cm tall. Each years growth was then 90% of the previous years growth. What is the maximum height of the wattle tree?

- (A) 490 cm
(B) 500 cm
(C) 570 cm
(D) 630 cm

Growth forms a geometric sequence with $a = 50$ and $r = 0.90$.

Limiting sum of a geometric series ($|r| < 1$)

$$S = \frac{a}{1-r}$$

$$= \frac{50}{1-0.9}$$

$$= 500 \text{ cm}$$

Maximum height of the tree is 570 cm or 5.7 m.

1 Mark: C

Q 2

**F
O
R
T
S
T**

Jack drops a super bouncy ball from the top of a 56 m building on to a concrete surface below. Its first rebound is 42 m, and each subsequent rebound is three quarters the height of the previous one.

- (i) How high will it rise on the fifth rebound?
(ii) How far will it travel in total?

(i)

Taking 42 as the first term as it is the first completed rebound.

$$T_n = ar^{n-1}$$

For the rise on the 5th bounce

$$a = 42 \quad r = \frac{3}{4} \quad n = 5$$

$$T_5 = 42 \left(\frac{3}{4} \right)^{5-1}$$

Note if take 56 as first term, need to find the 6th term.

$$T_5 = 42 \left(\frac{3}{4} \right)^4$$

$$\approx 13.29m$$

(ii) $|r| < 1 \quad r = \frac{3}{4}$

Consider one bounce up and down as a term, so $t_1 = 84$

$$\therefore S_{\infty} = \frac{a}{1-r}$$

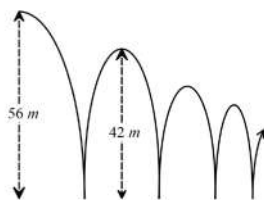
$$= \frac{84}{1-\frac{3}{4}}$$

$$= \frac{84}{\frac{1}{4}}$$

$$= 336m$$

Total distance travelled will be $336 + 56 = 392m$

Alternately take 42 as first term and double result from S_{∞}

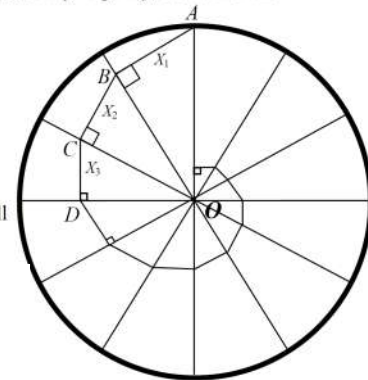


Q 3

CSSA

The diagram above shows a circular cartwheel with radius $OA = 4$ cm. The cartwheel is divided into 12 equal sectors. Perpendiculars $AB, BC, CD, \dots$ have been drawn and have respective lengths $X_1, X_2, X_3, \dots$ as shown.

- (i) By considering triangle OAB , show the distance $X_1 = 2$ cm.
(ii) Show that the distances $X_1, X_2, X_3, \dots$ form a geometric sequence.
(iii) Show that $X_{12} = 2 \left(\frac{\sqrt{3}}{2} \right)^{11}$ cm.
(iv) Calculate the length of one spiral, i.e. $(X_1 + X_2 + X_3 + \dots + X_{12})$.
(v) If the spiral is continued indefinitely, show that its total length will not exceed 15 cm.



(b) (i) (1mark)

Angle of each sector $= 360^\circ \div 12 = 30^\circ$. $\therefore$ In $\triangle OAB$, $\sin 30^\circ = \frac{X_1}{4}$ $\therefore \frac{1}{2} = \frac{X_1}{4}$

$$\therefore X_1 = 4 \times \frac{1}{2} = 2 \text{ cm.}$$

(b) (ii) (3marks)

Sample Answer

In $\triangle OBA$, $OB^2 = 4^2 - 2^2 = 12$ $\therefore OB = 2\sqrt{3}$ cm.

In $\triangle OCB$, $\sin 30^\circ = \frac{X_2}{OB} = \frac{X_2}{2\sqrt{3}}$ $\therefore \frac{1}{2} = \frac{X_2}{2\sqrt{3}}$ $\therefore X_2 = \sqrt{3}$

Similarly $X_3 = \frac{3}{2}$ $\therefore X_1, X_2, X_3, \dots = 2, \sqrt{3}, \frac{3}{2}, \dots$

$$\therefore \frac{\sqrt{3}}{2} = \frac{1}{2}\sqrt{3} \text{ and } \frac{3}{2} \div \sqrt{3} = \frac{1}{2}\sqrt{3}$$

$\therefore X_1, X_2, X_3, \dots$ forms a geometric sequence with a common ratio equal to $\frac{1}{2}\sqrt{3}$.

(b) (iii) (1mark)

Sample Answer

$$X_{12} = ar^{11} = 2 \times \left[\frac{\sqrt{3}}{2} \right]^{11}$$

(b) (iv) (2marks)

Sample Answer

$$S_{12} = \frac{2 \left[1 - \left(\frac{\sqrt{3}}{2} \right)^{12} \right]}{1 - \frac{\sqrt{3}}{2}} = \frac{2 \left[1 - \frac{729}{4096} \right]}{\frac{2-\sqrt{3}}{2}} = \frac{4 \times \frac{3367}{4096}}{2-\sqrt{3}} = \frac{3367}{1024} \div (2-\sqrt{3}) = 12.27 \text{ cm}$$

(b) (v) (1mark)

Sample Answer

$$S_{\infty} = \frac{2}{1 - \frac{\sqrt{3}}{2}} = \frac{4}{2-\sqrt{3}} \times \frac{2+\sqrt{3}}{2+\sqrt{3}} = 4(2+\sqrt{3}) = 14.9$$

$\therefore$ the total length will not exceed 15 cm.



series and sequence

Part D: GP Problems

Unit

NAME: _____

Q 4

GIRAWEEEN

A man undertook to pay \$200 to a charity one year, \$150 the next year, three-quarters of \$150 the third year and so on until he died. What is the greatest sum of money the charity may expect from these donations?

$$(a) \$200 + \$150 + \$\frac{3}{4} \times 150 + \dots$$

a geometric series with

$$a = \$200, r = \frac{3}{4}$$

Since $|r| < 1$, then

$$S = \frac{a}{1-r}$$

$$= \frac{\$200}{1 - \frac{3}{4}}$$

$$= \frac{\$200}{\frac{1}{4}}$$

$$= \$800$$

Q 5

HURLSTONE

A banksia bush was planted when it was 50cm tall. At the end of the first year after planting, the banksia was 90cm tall. Each year's growth was then 80% of the previous year's growth. What is the limiting maximum height of the banksia?

$$\text{Max. height} = 50 + \frac{40}{1 - 0.8} = 250 \text{ cm}$$

Q 6

HURLSTONE

The zoom function in a software package multiplies the dimensions of an image by 1.2. In an image, the height of the building is 50mm. After the zoom function is applied once, the height of the building in the image is 60mm. After a second application, its height is 72mm.

- Calculate the height of the building in the image after the zoom function has been applied eight times. Give your answer to the nearest mm.
- The height of the building in the image is required to be more than 400 mm. Starting from the original image, what is the least number of times the zoom function must be applied?

(d) (i)

$$60, 72, 86.4, 103.68, \dots$$

$$T_n = ar^{n-1}$$

$$T_8 = 60(1.2)^7$$

$$= 214.99$$

$$= 215 \text{ mm (nearest mm)}$$

(ii)

$$T_n = ar^{n-1}$$

$$T_n > 400$$

$$ar^{n-1} > 400$$

$$60(1.2)^{n-1} > 400$$

$$(1.2)^{n-1} > \frac{20}{3}$$

$$\log(1.2)^{n-1} > \log\left(\frac{20}{3}\right)$$

$$n-1 > \frac{\log\left(\frac{20}{3}\right)}{\log(1.2)}$$

$$n > 11.405$$

$$\therefore n = 12$$



series and sequence

Part D: GP Problems

2 unit

NAME: _____

Q 7

INDEPENDENT

Joe is trying to fill his 1 200 litre water tank.
The first day he puts 100 litres into the tank, the 2nd day he puts in 90 litres and the third day he puts in 81 litres.
If he continues to add water in this way will he ever fill the tank?
Justify your answer.

a) $S = 100 + 90 + 81 + \dots$
GP, where $a = 100$, $r = \frac{9}{10}$ limiting sum use $S_{\infty} = \frac{a}{1-r}$
$$S = \frac{100}{1 - \frac{9}{10}}$$
$$= 1000$$

No will never fill the tank as the limiting sum is only 1000 litres

Q 8

INDEPENDENT

Kevin has started an exercise program to lose weight. When he started the program he weighed 105 kg.
In the first month he lost 5 kg, in the second month he lost 4 kg and in the third month he lost 3.2 kg. If this weight loss trend continues

- how much will Kevin lose in the fourth month?
- what will be his ultimate weight?

c)(i) Outcome Assessed: H5

Answer:

5, 4, 3.2, ... $\therefore$ a GP.

$$r = \frac{3.2}{4} = \frac{4}{5}$$

$$T_4 = 3.2 \times \frac{4}{5}$$
$$= 2.56 \text{ kg}$$

c)(ii) Outcome Assessed: H5

Answer:

$$S_{\infty} = \frac{5}{1 - 0.8}$$
$$= 25 \text{ kg}$$

$$\text{Final weight } 105 - 25 = 80 \text{ kg}$$

Q 9

NEAP

A pile driver is hitting the top of a large pole. The first hit drives the pole 60 cm into the ground. The second hit drives the pole another $60 \times 0.75 = 45$ cm into the ground. The additional distance the pile goes into the ground with each drive is 75% of the previous distance.

- How far will the pole be driven into the ground on the 6th drive?
- Determine the total distance the pile will be driven on the 6th drive.
- Calculate the maximum distance the driver can drive the pole into the ground.

$$(i) T_6 = 60 \times (0.75)^5$$
$$= 14.24 \text{ cm}$$

$$(ii) S_6 = \frac{60(0.75^6 - 1)}{0.75 - 1}$$
$$= 197.29 \text{ cm}$$

$$(iii) S_{\infty} = \frac{60}{1 - 0.75}$$
$$= 240 \text{ cm}$$



series and sequence

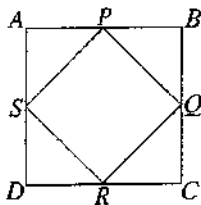
Part D: GP Problems

Unit

NAME: _____

Q10

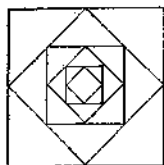
NEAP



$ABCD$ is a square with sides 16 cm long. P , Q , R and S are the midpoints of the sides of the square $ABCD$. P , Q , R and S are joined to make another square.

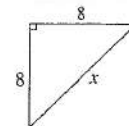
(i) Show that $PS = 8\sqrt{2}$ and that the area of $PQRS$ is 128 cm^2 .

A 'squares within squares' pattern is produced by joining midpoints of the sides of successive squares.



- (ii) $ABCD$ is the first square and $PQRS$ is the second square. What is the area of the 10th square?
- (iii) Which square has a perimeter of $\sqrt{2} \text{ cm}$?
- (iv) Imagine the pattern can be repeated infinitely. What is the relationship between the sum of the areas of all the squares and the original square $ABCD$? Use a calculation to justify your answer.

(i)



$$x^2 = 64 + 64$$

$$= 64 \times 2$$

$$x = 8\sqrt{2}$$

$$\begin{aligned} \text{area of } PQRS &= (8\sqrt{2})^2 \\ &= 128 \text{ cm}^2 \end{aligned}$$

(ii) The areas are 256, 128, ...

This is a geometric sequence: $a = 256$, $r = \frac{1}{2}$

$$\begin{aligned} T_{10} &= ar^9 \\ &= 256 \times \left(\frac{1}{2}\right)^9 \\ &= \frac{1}{2} \text{ cm}^2 \end{aligned}$$

(iii) The perimeters are 64, $32\sqrt{2}$, $32 \dots$

$$T_n = \sqrt{2} = 64 \times \left(\frac{1}{\sqrt{2}}\right)^{n-1}$$

$$(\sqrt{2})^n = 64$$

$$2^{\frac{1}{2}n} = 2^6$$

$$n = 12$$

(iv) areas = 256, 128, ..., $r = \frac{1}{2}$

$$S_{\infty} = \frac{a}{1 - \frac{1}{2}}$$

$$= 2a$$

The sum of the areas of all the squares is twice the area of the original square.



series and sequence

Part D: GP Problems

2unit

NAME: _____

Q11

NSBHS

In order to reduce the 'bill shock' that occurs whenever he receives his mobile phone bill, Mr Lam decides to reduce his mobile data usage by 15% of the previous day's usage, starting from the first day of the billing cycle.

On the first day, he used 200 megabytes (MB) of mobile data.

- Find the amount of data (in MB) used on the fifth day, correct to 2 decimal places.
- Find the total amount of data (in MB) used after 7 days, correct to 2 decimal places.
- Show that a mobile phone plan which offers 1500MB of data per month would be sufficient for his usage.

(a) i. (1 mark)

$$\begin{aligned} a &= 200 & r &= 0.85 \\ T_5 &= ar^{n-1} \\ &= 200(0.85)^4 \\ &= 104.4 \text{ MB} \end{aligned}$$

ii. (2 marks)

$$\begin{aligned} S_7 &= \frac{a(r^n - 1)}{r - 1} \\ &= \frac{200(0.85^7 - 1)}{0.85 - 1} \\ &= 905.9 \text{ MB} \end{aligned}$$

iii. (2 marks)

- ✓ [1] for significant progress towards answer.
✓ [1] for final answer.

$$\begin{aligned} S &= \frac{a}{1 - r} \\ &= \frac{200}{1 - 0.85} = 1333.33 \text{ MB} \end{aligned}$$

Maximum data usage would be 1333.33 MB. Hence a 1500 MB data plan would be sufficient.

Q12

NSGHS

Rapunzel has discovered something interesting about the way her hair grows.

Initially her hair is 18 metres long.

She measures her hair every year and records how much her hair grows during

each year, **10 m** **9 m** **8.1 m** **7.29 m**

- Find how much Rapunzel's hair will grow during 10th year, correct to 2 decimal places
- Find the length of Rapunzel's hair after ten years, correct to 2 decimal places.
- If the tower is 110 metres high, assuming that her hair continues to grow in a similar pattern, will Rapunzel's hair ever be long enough to reach the ground? Justify your answer.

$$(e) \text{ (i) } r = \frac{9}{10} = \frac{8.1}{9} = 0.9$$

The sequence is GP
with $a = 10$ and $r = 0.9$

$$\begin{aligned} T_{10} &= ar^9 \\ &= 10 \times (0.9)^9 \\ &= 3.87 \end{aligned}$$

∴ 3.87 metres

$$\begin{aligned} \text{(ii) } S_{10} &= \frac{a(1 - r^{10})}{1 - r} \\ &= \frac{10(1 - 0.9^{10})}{1 - 0.9} \\ &= 65.13 \end{aligned}$$

Total length of her hair is 83.13 m

$$\text{(iii) } r = 0.9 < 1 \quad \therefore S_{\infty} \text{ exists}$$

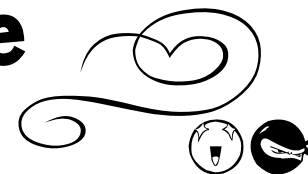
$$\begin{aligned} S_{\infty} &= \frac{a}{1 - r} \\ &= \frac{10}{1 - 0.9} \\ &= 100 \end{aligned}$$

The max length of her hair is 100 metres.
Yes it will be long enough to reach the ground.



series and sequence

Part e: misc.



Unit

NAME:

Q 1

ACE

The sum of the firsts n terms of a certain arithmetic series is given by:

$$S_n = \frac{3n^2 + n}{2}$$

- Calculate S_1 and S_2 .
- Find the first three terms of the series.
- Find an expression for the n th term.

1
2
1

13(a) (i)	$S_n = \frac{3n^2 + n}{2}$ $S_1 = \frac{3 \times 1^2 + 1}{2} = 2$	$S_n = \frac{3n^2 + n}{2}$ $S_2 = \frac{3 \times 2^2 + 2}{2} = 7$	1 Mark: Correct answer.
13(a) (ii)	$a = T_1 = S_1 = 2$ $T_2 = S_2 - S_1 = 7 - 2 = 5$ $d = T_2 - T_1 = 5 - 2 = 3$ Sequence is $\{2, 5, 8, \dots\}$		2 Marks: Correct answer. 1 Mark: Finds the first term or the common difference.
13(a) (iii)	$T_n = a + (n-1)d$ $= 2 + (n-1)3$ $= 2 + 3n - 3$ $= 3n - 1$		1 Mark: Correct answer.

Q 2

BAULKHAM

The sum of a series is given as $S_n = n^2 - 3^n$.

- Find S_{12} , leaving your answer in 2 decimal places.
- Find the 12th term of the series in 2 decimal places.

1
2

(i) $S_{12} = 12^2 - 3^{12}$
 $= -531297 \checkmark$

(ii) $T_{12} = S_{12} - S_{11}$
 $S_{11} = 11^2 - (3)^{11}$
 $= -177026 \checkmark$
 $\therefore T_{12} = -531297 - (-177026)$
 $= -354271 \checkmark$



series and sequence

Part e: misc.

Unit

NAME: _____

Q 3

CSSA

With the drought ever worsening, James and Theodore design a counting generator that can simulate the number of rain drops per minute that fall over a river during a storm.

The rain drops falling per minute forms the series

$$1+1+3+9+23+\dots$$

with the n th term given by the formula $R_n = 1 - 2n + 2^n$.



- Verify that 115 is a term of this series.
- Find the total amount of rain drops which fall over the river in the first twenty five minutes.
- If the surface area of the river is 250 m^2 find the average rainfall per cm^2 over the first twenty five minutes.

$$R_6 = 53$$

$$R_7 = 115$$

$$\therefore n = 7, R = 115$$

(b) (ii) (2 marks)

Sample Answer

$$1+1+3+9+23+\dots = (-1+2)+(-3+4)+(-5+8)+\dots(-49+33\ 554\ 432)$$

$$AP: S_{25} = \frac{25}{2}(-1+49) = -625$$

$$GP: S_{25} = \frac{2(2^{25}-1)}{2-1} = 67\ 108\ 862$$

After 25 mins we have $-625 + 67\ 108\ 862 = 67\ 108\ 237$ rain drops

(b) (iii) (1 mark)

Sample Answer

$$\text{Raindrops per cm}^2 = 67\ 108\ 237 \div 2\ 500\ 000 = 26.8 \text{ drops/cm}^2$$

Q 4

GIRAWEEEN

Evaluate $\sum_{k=2}^5 \frac{(-1)^k}{k+1}$

$$\begin{aligned} (c) \sum_{k=2}^5 \frac{(-1)^k}{k+1} \\ &= \frac{(-1)^2}{2+1} + \frac{(-1)^3}{3+1} + \frac{(-1)^4}{4+1} + \frac{(-1)^5}{5+1} \\ &= \frac{1}{3} - \frac{1}{4} + \frac{1}{5} - \frac{1}{6} \\ &= \frac{7}{60} \quad \# \quad (2) \end{aligned}$$

Q 5

GIRAWEEEN

The numbers p , q and r add to 9 and form an arithmetic progression. The numbers r , p and q form a geometric progression. Find the values of p , q and r .

(c) p, q, r form an AP

$$\therefore q - p = r - q$$

$$2q = p + r$$

Also $q + p + r = 9$

$$q + 2q = 9$$

$$3q = 9$$

$$q = 3 \dots (1)$$

and $p + r = 6 \dots (2)$

r, p, q form a GP

$$\frac{p}{r} = \frac{q}{p}$$

$$p^2 = r q$$

Using (1) & (2)

$$p^2 = (6-p)3$$

$$p^2 = 18 - 3p$$

$$p^2 + 3p - 18 = 0$$

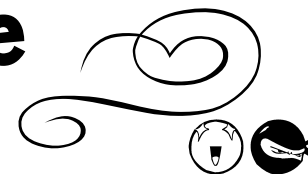
$$(p+6)(p-3) = 0$$

$$\begin{cases} p = -6 \\ q = 3 \\ r = 12 \end{cases} \quad \begin{cases} p = 3 \\ q = 3 \\ r = 3 \end{cases}$$



series and sequence

Part e: misc.



2 unit

NAME: _____

Q 6

INDEPENDENT

The sum of the first n terms of an arithmetic series is given by: $S_n = n(38 - 2n)$

- Calculate S_1 and S_2 .
- Find the first three terms of this series.
- Find an expression for the n -th term.

(i)	$S_n = n(38 - 2n)$	(ii)	$T_2 = S_2 - S_1$
	$S_1 = 1(38 - 2)$		$= 68 - 36$
	$= 36$		$= 32$
	$S_2 = 2(38 - 4)$		$d = -4$
	$= 2 \times 34$		$T_3 = 28$
	$= 68$		$36, 32, 28, \dots$
		(iii)	$T_n = 36 + (n-1)(-4)$
			$= 36 - 4n + 4$
			$= 40 - 4n$

Q 7

INDEPENDENT

Jeff and Barry have been offered jobs in a mine. The jobs are for 20 days.

Jeff agrees to be paid 5 cents for the 1st day's work, 10 cents for the 2nd day's work, 20 cents for the 3rd day's work with the company doubling his wage each day of the 20 day contract.

Barry agrees to be paid \$500 for the 1st day's work, \$550 for the 2nd day's work, \$600 for the 3rd day's work with the company increasing his wage by \$50 a day for each day of the 20 day contract.

- How much will Jeff earn on the 16th day of his contract?
- How much will Barry earn on the 16th day of his contract?
- Barry finds the job too hard and only completes 10 days. Show that he earns \$7250 for the 10 day's work.
- What is the minimum number of days Jeff will have to work to ensure that he earns more than Barry's wage of \$7250?

c)(i) Outcome Assessed: H5

Answer:

0.05, 0.10, 0.20, 0.40, 0.80,

$$a = 0.05, \quad r = 2, \quad T_n = ar^{n-1}$$

$$T_{16} = (0.05) \times 2^{15}$$

$$= \$1638.40$$

c)(ii) Outcome Assessed: H5

Answer:

500, 550, 600, 650,

$$a = 500, \quad d = 50, \quad T_n = a + (n-1)d$$

$$T_{16} = 500 + 15 \times 50$$

$$= \$1250$$

c)(iii) Outcome Assessed: H5

Answer:

$$S_n = \frac{n}{2}(2a + (n-1)d)$$

$$S_{10} = 5(1000 + 9 \times 50)$$

$$= \$7250$$

c)(iv) Outcome Assessed: H5

Answer:

$$S_n = \frac{a(r^n - 1)}{r - 1}$$

$$S_n = \frac{0.05(2^n - 1)}{1}$$

$$0.05(2^n - 1) > 7250$$

$$2^n - 1 > 145000$$

$$2^n > 145001$$

$$n \ln(2) > \ln(145001)$$

$$n > \frac{\ln(145001)}{\ln(2)}$$

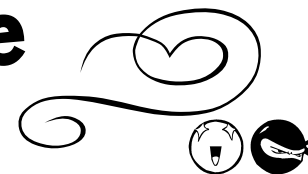
$$n > 17.1457$$

$$18 \text{ days}$$



series and sequence

part e: misc.



2unit

NAME: _____

Q 8

NORTH SYDNEY BOYS

The sum to n terms of a sequence of numbers is given by $S_n = 102n - 2n^2$.

- Find an expression for T_n , the n -th term of the sequence.
- What type of a sequence is this?

i. (2 marks)

$$S_n = 102n - 2n^2$$

$$T_n = S_n - S_{n-1}$$

$$= 102n - 2n^2$$

$$- (102(n-1) - 2(n-1)^2)$$

$$= 102n - 2n^2$$

$$- (102n - 102 - 2(n^2 - 2n + 1))$$

$$= 102n - 2n^2 + 102 - 2n^2 + 4n - 2$$

$$= 104 - 4n$$

ii. (1 mark)

$$T_1 = 104 - 4(1) = 100$$

$$T_2 = 104 - 4(2) = 96$$

$$T_3 = 104 - 4(3) = 92$$

$$T_3 - T_2 = T_2 - T_1$$

Arithmetic sequence.